

College Algebra Skills Hub Overview

A shared resource for review practice and targeted support

Overview

The College Algebra Skills Hub would give students one dependable place to review prerequisite algebra, revisit important College Algebra concepts, practice skills, and receive additional instruction when they need it. Students currently have access to many outside resources, but those resources can be difficult to locate and may use methods, notation, or expectations that differ from what is taught in class. The Hub would provide an organized resource that instructors and tutors could recommend with confidence.

D2L would serve as the main home for the Hub. It would contain the complete collection of skill pages, notes, worksheets, videos, animations, practice, tutorials, and links. A companion Math Portal could provide easier access to interactive activities from a phone, tablet, or computer. Together, these tools would support students who need a quick reminder, additional practice, guided help, or a more complete explanation.

This overview describes the intended system without setting a production schedule or defining a final list of skills. Those decisions would be made during the next planning stage after current materials have been reviewed and faculty priorities have been discussed.

Core Parts of the Hub

Part	Purpose
D2L Skills Hub	The organized home for the complete collection and the main entry point for students
Instructional resources	Short animations, notes, examples, worksheets, and longer lessons that explain important skills
Independent practice	Quiz-style practice that allows students to work on their own and receive immediate correctness feedback
Guided practice and tutorials	Interactive activities that help students through a process, respond to errors, and provide more complete instruction
Math Portal	A companion web application for interactive work, progress saving, reporting, surveys, and convenient device access
Instructor-specific options	Selected alternatives for topics where notation, required steps, or teaching methods differ between instructors
Tutoring integration	A shared skill map and direct links that tutors can use to identify missing skills and guide continued practice

Student Experience

Students would be able to enter the Hub from D2L, a direct instructor link, a tutoring recommendation, or eventually the Math Portal. They could locate a skill by course topic, prerequisite area, or common task and then choose the amount of support they need. A student who has forgotten one rule might use a short animation and worked example. A student who needs practice could complete an independent activity. A student who is struggling with the process could move into guided practice or a full tutorial.

Students would not be expected to complete every resource. The goal is to make the appropriate help easy to find and to give instructors and tutors a clear place to send students for targeted review. When useful, activity data and student feedback could also show where students are struggling and where an explanation or application should be improved.

Content and Organization

The final content scope has not yet been determined. The Hub would include prerequisite skills that students are expected to know and selected College Algebra concepts that benefit from additional explanation or practice. The next planning stage would create a skill map, review the materials already available in the Boot Camp and current applications, and identify the first topics to develop. The Hub is not intended to reproduce every part of the College Algebra course or every feature of a commercial homework system.

Organization will be essential as the collection grows. Students should be able to browse, search using familiar language, or follow a direct link without needing to know the formal name of a missing skill. Skill pages should use a consistent structure so explanations, examples, practice, guided help, and related skills are easy to recognize. The detailed navigation and naming system would be designed as part of the implementation plan.

Development Approach

The Hub would be developed in phases rather than built all at once. Early work would focus on organizing what already exists, establishing the skill structure, and creating resources that can be produced and reused efficiently. Short instructional animations and concise reference materials could be expanded first. Independent practice, guided activities, and full tutorials could then be added where the instructional need justifies the additional development time.

Later phases could expand Math Portal access, add instructor-specific versions, and strengthen coordination with tutoring staff and other faculty contributors. Classroom use, student feedback, and tutor or instructor observations would help determine which resources should be revised or developed next. The scope of each phase would be set only after the initial inventory and priority decisions are complete.

Next Planning Stage

Before a detailed implementation proposal is created, the following decisions need to be made:

- Inventory the Boot Camp, existing animations, current applications, and other available materials.
- Create the overall skill map and distinguish prerequisite review from concepts taught within College Algebra.

- Select the first skills and resource types to develop based on student need, instructional importance, and available materials.
- Design the basic D2L organization, naming system, navigation, and connection to the Math Portal.
- Determine how instructors and tutoring staff will review, recommend, and use the resources.
- Define a realistic first phase and then identify the time, responsibilities, technology, and support needed to complete it.

The result of that planning work would be a separate implementation scope with clear priorities, phases, responsibilities, and resource needs. This overview can be used first to explain the concept and gather feedback on what the Hub should become.